

1) Define Reinforcement Learning (RL). Explain its main objective.

A) Reinforcement Learning (RL) is a type of machine Learning in which an agent learns by interacting with an environment and receives rewards or penalties for its actions.

Main objective:-

- Learn the best policy (strategy).
- Maximize the total cumulative reward over time.

2) What is a Markov Decision process (MDP)? List its components.

A) A Markov Decision process (MDP) is a mathematical model used in RL to describe decision making where the next state depends only on the current state and action.

Components:-

- 1. S - set of states

2. A - Set of Actions

3. p - Transition probability

4. R - Reward Function

5. γ (Gamma) - Discount Factor

3) State the Bellman Equation for the state-value function.

A) The Bellman Equation is

$$v(s) \leq_a \pi(a|s) \sum_{s'} p(s'|s,a) [R(s,a,s') + \gamma v(s')]$$

Where :

• $v(s)$ = state value

• $\pi(a|s)$ = policy

• γ = Discount factor

4) Differentiate between Exploration and Exploitation.

Exploration

- Tries new actions
- Gains more knowledge
- may give reward
- Helps discover better actions

Exploitation

- uses known best action
- maximizes immediate reward
- usually gives higher reward
- uses existing knowledge

5) What is the multi-Armed Bandit problem?

A) The multi-Armed Bandit problem is a decision making problem where an agent chooses one among several actions (slot machines) with unknown

rewards to maximize the total reward.

Goal: Balance exploration and exploitation.

6) What is Dynamic programming (DP)? Mention its applications in RL.

A) Dynamic programming (DP) is a method used to solve complex problems by breaking them into smaller subproblems.

Applications in RL:

- Policy evaluation
- Policy improvement
- Policy iteration
- Value iteration
- Finding the optimal policy.

7) What is Temporal Difference (TD) Learning? Write the TD(0) update equation.

A) Temporal Difference (TD) Learning is an RL method that learns by updating value estimates using the difference between predicted and actual rewards.

TD(0) update equation:

$$V(s) = V(s) + \alpha [R + \gamma V(s') - V(s)]$$

Where:

- α = learning rate, γ = discount factor
- R = Reward, s' = Next state

8) Differentiate between SARSA and Q-learning.

SARSA

- on-policy algorithm
- uses actions actually taken
- safer learning
- more conservative

Q-learning

- off-policy algorithm
- uses best possible action.
- Faster learning
- more aggressive

9) What is a Deep Q-Network (DQN)? Mention its advantages over tabular Q-learning.

A) A Deep Q-Network (DQN) uses a deep neural network to estimate Q-values instead of a Q-table.

Advantages:

- Handles large state spaces.
- Better generalization.

10) What is Double Deep Q-Network (DDQN)? Why is it preferred over DQN?

A) A Double Deep Q-Network (DDQN) is an improved version of DQN that uses two networks to reduce overestimation of Q-values.

Why preferred over DQN?

- Reduces overestimation bias.
- more stable learning.